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Eurobioc2026 BiocContainer Report

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Introduction

Containers enhance reproducibility and additionally offer a convenient way to share the setup used and needed to run a script. Offering a way to create containers which contain desired `R` packages from within `R` allows users to more easily create these containers and work with them. `R` packages like dockerfiler Fay et al. (2026) offer functions to create and work with Docker files within `R`. However, currently there is no convenient way to create Docker files and containers from `R` while also controlling for the bioconductor release. The aim in this hackathon was to create a minimal set of functions that offer this possibility, and this work eventually gave rise to the R/Bioconductor package `SimpleBiocContainer`.

R/Bioconductor package `SimpleBiocContainer`

As part of the European Bioconductor Hackathon 2026, we created an R/Bioconductor package called `SimpleBiocContainer` which offers a more user friendly way to create, build and push Docker containers to Dockerhub from R. A working version of the package can be found on GitHub: (https://github.com/lgatto/SimpleBiocContainer.git).

The functions in `SimpleBiocContainer` make use of the existing Bioconductor docker containers which already contain a few packages, and additionally add the user-specified Bioconductor/R packages using `BiocManager::install()`. The Bioconductor version is inferred from the R session and the corresponding docker container matching that release is then used.

ToDo: describe the functions and what they do, maybe illustrate an example run. link to GH pacge of the package. I avoided naming the `makeSimpleBiocContainer` function for now, should this be explicitly mentioned or is it better not to be too detailed in case things change in the near future.

Conclusions / Future Work

get list of R packages loaded in session to create a container for those. add vignettes, further enhacncements.

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Fay, C., Guyader, V., Parry, J., & Rochette, S. (2026). *Dockerfiler: Easy dockerfile creation from r*. https://doi.org/10.32614/CRAN.package.dockerfiler

Author contributions